FAST- National University of Computer and Emerging Sciences, Karachi.

FAST School of Computing

Assignment # 2, Spring 2026

CS3001- Computer Networks

ASSIGNMENT-II (DNS Implementation)

Submission Guidelines:

- This is an individual assignment.
- This is a programming assignment and the submission will be based on 4-5 minutes demo followed by a short viva on Monday 06th April, 2026.
- You must submit the code and screenshot of the working demo on Google classroom.
- You must mark the attendance during the demo. The student ID, name and section must be mentioned.
- Plagiarism in any form will result in straight "F".

Assignment # 2 (100 points)

Suppose within your web browser you click on a link to obtain a Web Page. The requested IP Address is not cached in your local host, so a DNS lookup is necessary to obtain the IP Address, refer to Figure 01.

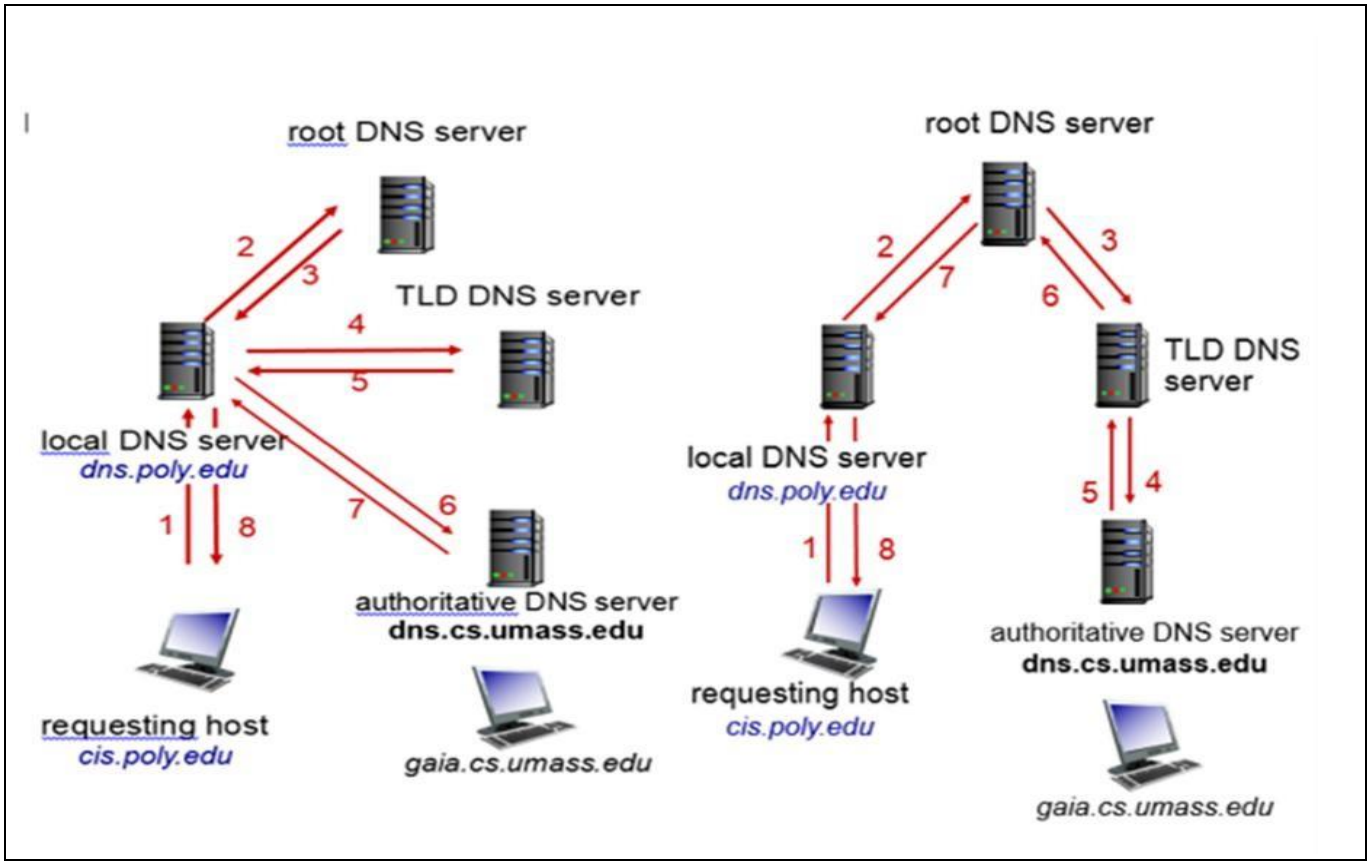

Figure 01: DNS name resolution (The right side of figure is showing Iterative name resolution and leftside of figure is showing sequential name resolution)

Your task:

- **Design an application for DNS-Server which implements the Root-Level, Top-Level and Authoritative-Level servers, show how these servers actually work. When the client send query for DNS lookup, it query and reply messages, both with same message format. It take16# bit for identification and 16# bit for reply to query, refer to Figure 02.**

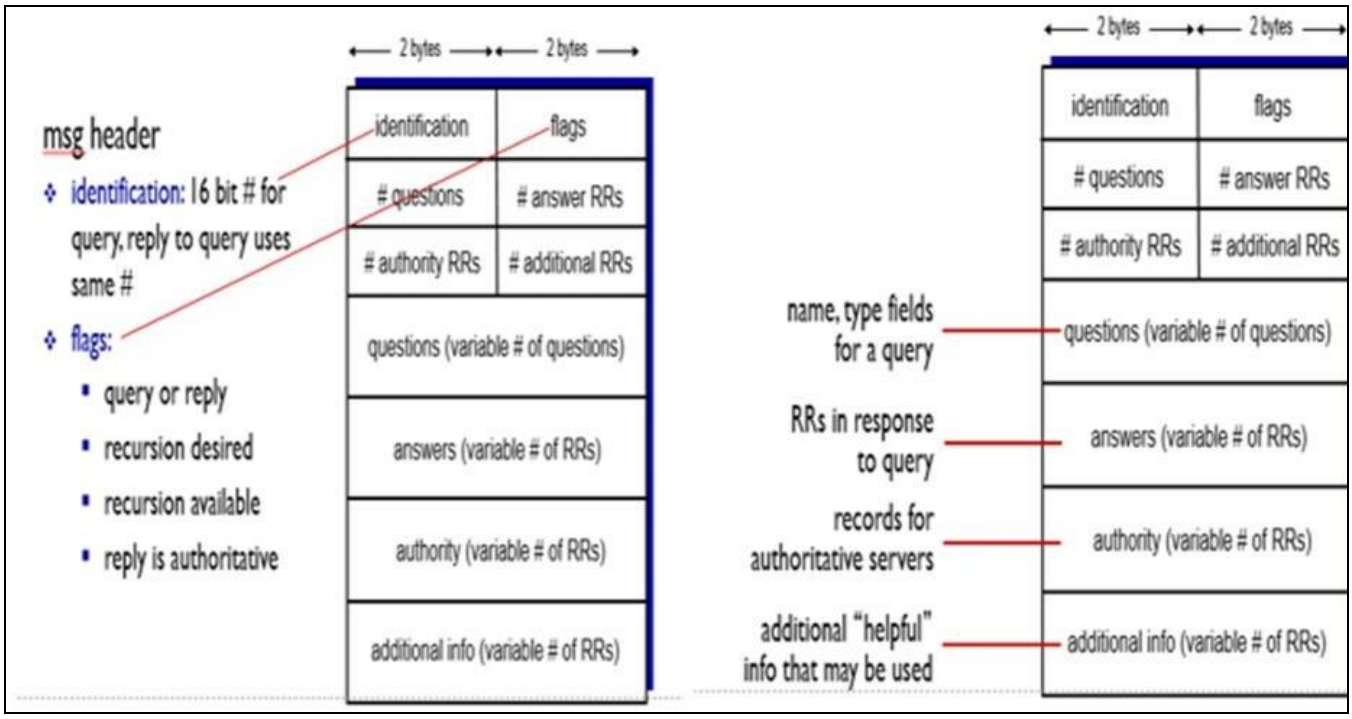

Figure 02: DNS, protocol messages

- **Your application should maintain DNS record list (as per actual records - no assumption), which prints requested IP Address records.**

```
Sample run for google.com:
google.com/64.233.167.99
-- DNS INFORMATION --
A: 64.233.187.99, 72.14.207.99, 64.233.167.99
NS: ns4.google.com., ns1.google.com., ns2.google.com.,
ns3.google.com.
MX: 10 smtp4.google.com., 10 smtp1.google.com., 10
smtp2.google.com., 10smtp3.google.com.
```

- **Show how caching a DNS request will help for recursive request.**
- **Enable auto flushing of records if cache gets full on local host level.**

Help:

Reading: Text-Book Chapter # 02 (Section 2.5)

http://www.eyeasme.com/Shayne/DNSLOOKUP/DNSLookup.html